

LAB WORK 1

Part 1. Key Identification Exercises

Task 1.1 — Relation A: Employee

Employee(EmpID, SSN, Email, Phone, Name, Department, Salary)

EmpID	SSN	Email	Phone	Name	Department	Salary
101	123-45-6789	john@company.com	555-0101	John	IT	75000
102	987-65-4321	mary@company.com	555-0102	Mary	HR	68000
103	456-78-9123	bob@company.com	555-0103	Bob	IT	72000

1. Superkeys

- {EmpID}
- {SSN}
- {Email}
- {Phone}
- {EmpID, Name}
- {EmpID, Department}
- {SSN, Name}
- {Email, Phone}

2. Candidate keys

The candidate keys are {EmpID}, {SSN}, {Email}, {Phone}. A candidate key is a minimal superkey.

3. Primary key choice

EmpID is the best primary key because it is short, stable, specifically created for employee identification, and does not expose sensitive information. Email and phone may change.

4. Phone number

The sample shows unique phone numbers. However, Phone is a candidate key only if a business rule guarantees uniqueness; sample data alone is not enough.

Task 1.1 — Relation B: Course Registration

Registration(StudentID, CourseCode, Section, Semester, Year, Grade, Credits)

Primary key: (StudentID, CourseCode, Section, Semester, Year)

- StudentID identifies the student.
- CourseCode identifies the course.
- Section distinguishes sections of the same course.
- Semester is needed because a course may be taken in different semesters.
- Year distinguishes Fall 2025 from Fall 2026.

No additional candidate key is specified by the given business rules.

Task 1.2 — Foreign Key Design

Foreign key	References
Student.AdvisorID	Professor.ProfID
Professor.Department	Department.DeptCode

Foreign key	References
Course.DepartmentCode	Department.DeptCode
Department.ChairID	Professor.ProfID
Enrollment.StudentID	Student.StudentID
Enrollment.CourseID	Course.CourseID

Part 2. ER Diagram Construction

Task 2.1 — Hospital Management System

Entities

- Patient — strong entity.
- Doctor — strong entity.
- Department — strong entity.
- Appointment — entity recording patient-doctor appointments.
- Prescription — entity recording prescriptions.
- HospitalRoom — weak entity because RoomNumber is unique only within a department.

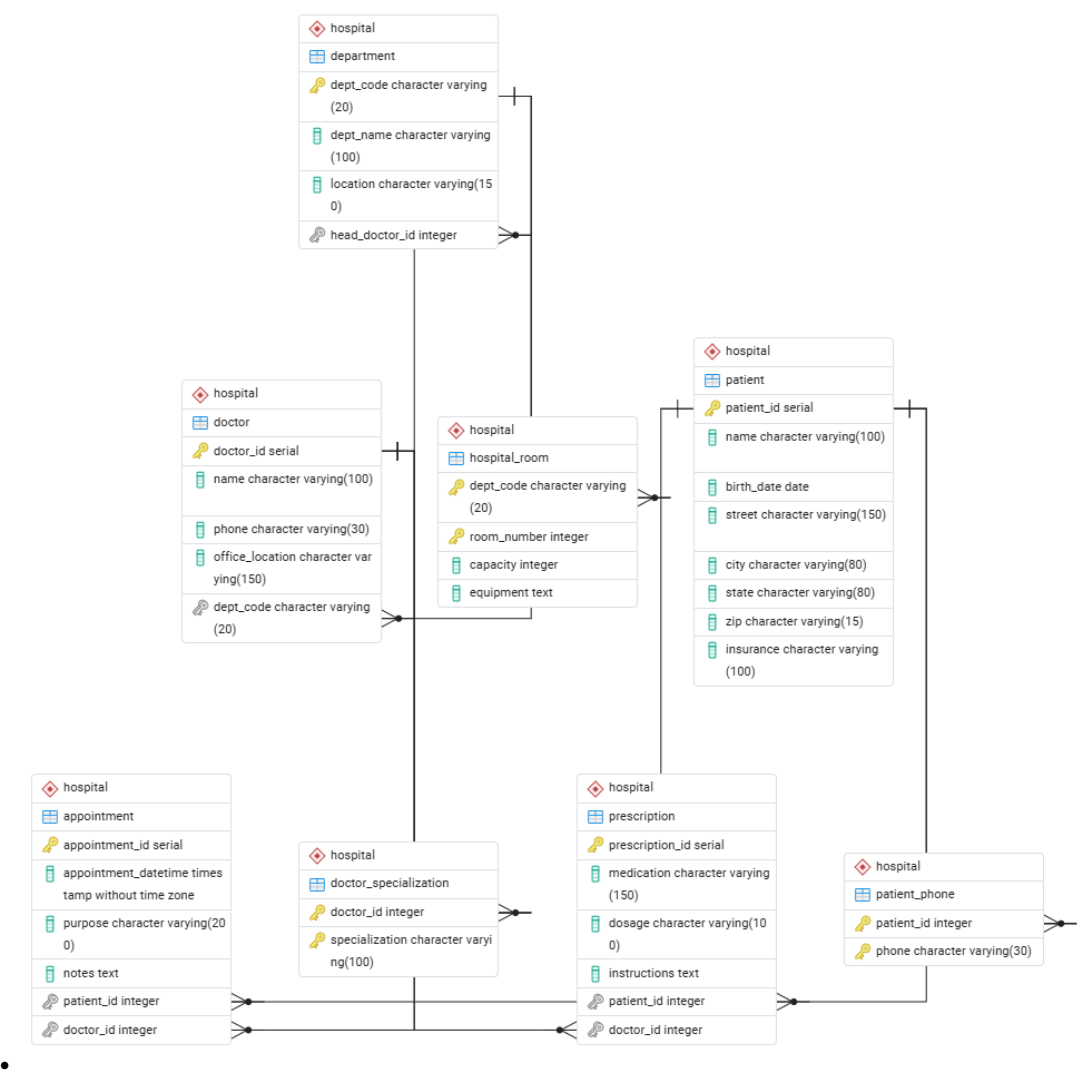
Attributes

Entity	Attributes / classification
Patient	PatientID PK, Name, BirthDate, Address (composite: Street, City, State, ZIP), Phone (multivalued), Insurance; Age can be de
Doctor	DoctorID PK, Name, Specialization (multivalued), Phone (multivalued), OfficeLocation.
Department	DeptCode PK, DeptName, Location.
Appointment	AppointmentID PK, DateTime, Purpose, Notes, PatientID FK, DoctorID FK.
Prescription	PrescriptionID PK, Medication, Dosage, Instructions, PatientID FK, DoctorID FK.
HospitalRoom	DeptCode PK/FK, RoomNumber partial key; full key = (DeptCode, RoomNumber).

Relationships / cardinalities

- Patient 1:N Appointment
- Doctor 1:N Appointment
- Patient M:N Doctor through Appointment
- Patient 1:N Prescription
- Doctor 1:N Prescription

- Department 1:N HospitalRoom (identifying relationship).

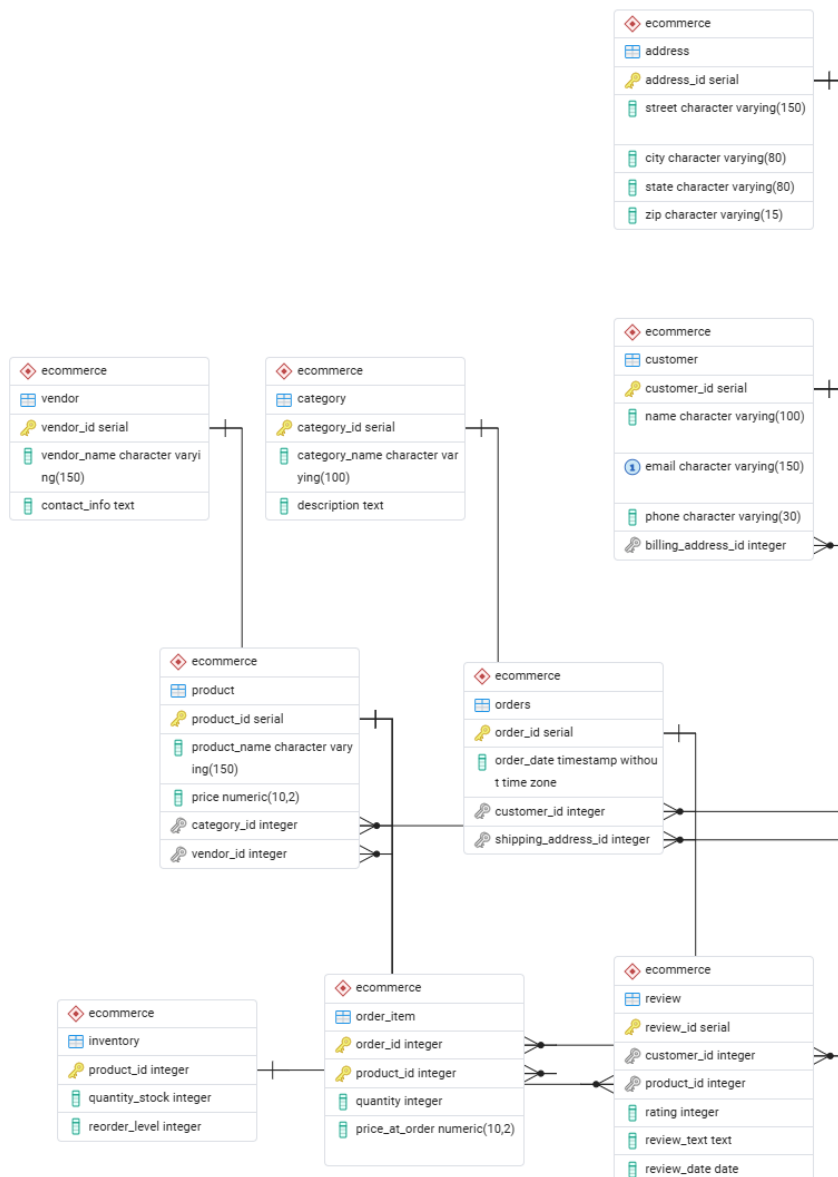


Task 2.2 — E-commerce Platform

Entity	Main attributes
Customer	CustomerID PK, Name, Email, BillingAddressID FK
Order	OrderID PK, OrderDate, CustomerID FK, ShippingAddressID FK
OrderItem	OrderID PK/FK, ProductID PK/FK, Quantity, PriceAtOrder
Entity	Main attributes
Product	ProductID PK, ProductName, Price, CategoryID FK, VendorID FK
Category	CategoryID PK, CategoryName
Vendor	VendorID PK, VendorName
Review	ReviewID PK, CustomerID FK, ProductID FK, Rating, ReviewText, ReviewDate
Inventory	ProductID PK/FK, QuantityInStock
Address	AddressID PK, Street, City, State, ZIP

Relationships

- Customer 1:N Order
- Order 1:N OrderItem
- Product 1:N OrderItem
- Order M:N Product through OrderItem; Quantity and PriceAtOrder are relationship attributes.
- Category 1:N Product
- Vendor 1:N Product
- Customer 1:N Review • Product 1:N Review
- Product 1:1 Inventory
- Customer 1:N Address; an order stores the selected shipping address.



Part 4. Normalization Workshop

Task 4.1 — StudentProject

StudentProject(StudentID, StudentName, StudentMajor, ProjectID, ProjectTitle, ProjectType, SupervisorID, SupervisorName, SupervisorDept, Role, HoursWorked, StartDate, EndDate)

Functional dependencies

- StudentID → StudentName, StudentMajor
- ProjectID → ProjectTitle, ProjectType, SupervisorID
- SupervisorID → SupervisorName, SupervisorDept
- (StudentID, ProjectID) → Role, HoursWorked, StartDate, EndDate

Anomalies

Problem	Explanation
Redundancy	Project and supervisor information is repeated for every student assigned to a project.
Update anomaly	Changing supervisor data requires updates in multiple rows.
Insert anomaly	A project cannot be represented naturally if no student is assigned yet.
Delete anomaly	Deleting the last student-project row may remove the only stored project/supervisor information.

1NF

The relation is already in 1NF assuming every attribute contains an atomic value and there are no repeating groups.

2NF

Primary key = (StudentID, ProjectID). Student attributes depend only on StudentID, while project attributes depend only on ProjectID, creating partial dependencies.

STUDENT(StudentID PK, StudentName, StudentMajor)

PROJECT(ProjectID PK, ProjectTitle, ProjectType,
SupervisorID, SupervisorName, SupervisorDept)

STUDENT_PROJECT(StudentID PK/FK, ProjectID PK/FK,
Role, HoursWorked, StartDate, EndDate)

3NF

Transitive dependency: ProjectID → SupervisorID → SupervisorName, SupervisorDept. Therefore supervisor information is separated.

STUDENT(StudentID PK, StudentName, StudentMajor)

SUPERVISOR(SupervisorID PK, SupervisorName, SupervisorDept)

PROJECT(ProjectID PK, ProjectTitle, ProjectType, SupervisorID FK)

STUDENT_PROJECT(StudentID PK/FK, ProjectID PK/FK,
Role, HoursWorked, StartDate, EndDate)

Task 4.2 — Advanced Normalization

CourseSchedule(StudentID, StudentMajor, CourseID, CourseName, InstructorID, InstructorName, TimeSlot, Room, Building)

Assumption: Since SectionID is absent, we interpret (CourseID, TimeSlot) as identifying a course section. If that is not true, SectionID should be added.

Primary key: (StudentID, CourseID, TimeSlot)

Functional dependencies

- StudentID → StudentMajor
- CourseID → CourseName
- InstructorID → InstructorName
- Room → Building
- (CourseID, TimeSlot) → InstructorID, Room

BCNF check

The original relation is not in BCNF because determinants such as StudentID, CourseID, InstructorID and Room are not candidate keys of the whole relation.

BCNF decomposition

STUDENT(StudentID PK, StudentMajor)

COURSE(CourseID PK, CourseName)

INSTRUCTOR(InstructorID PK, InstructorName)

ROOM(Room PK, Building)

COURSE_SECTION(CourseID PK/FK, TimeSlot PK, InstructorID FK, Room FK) ENROLLMENT(StudentID PK/FK, CourseID PK/FK, TimeSlot PK/FK)

Under the stated assumptions the decomposition is intended to be lossless. If multiple sections of the same course can share a time slot, the original schema is ambiguous and a SectionID is recommended.

Part 5. Design Challenge

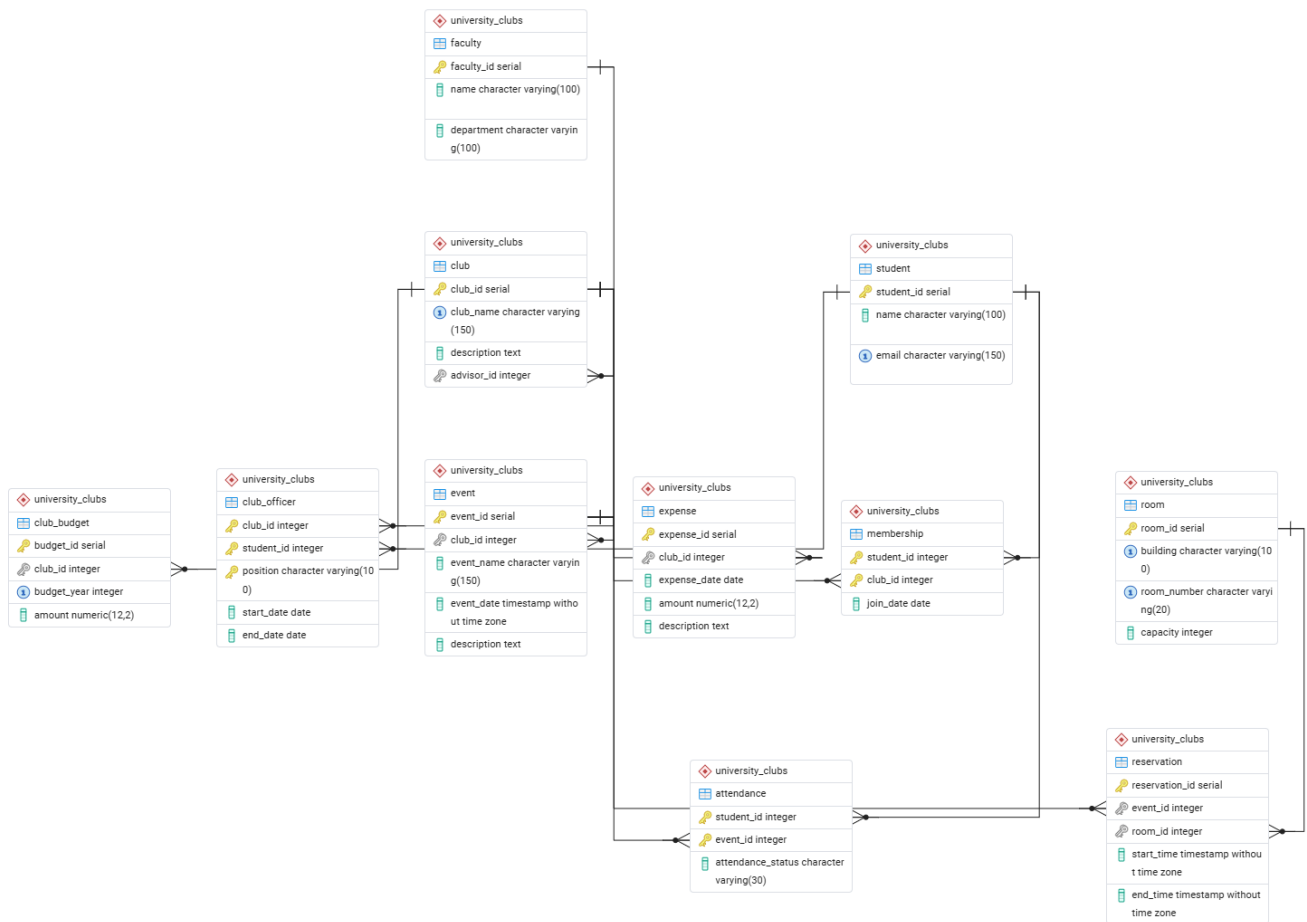
Task 5.1 — University Student Clubs and Organizations

Normalized relational schema

Table	Schema
STUDENT	StudentID PK, Name, Email
FACULTY	FacultyID PK, Name, Department
CLUB	ClubID PK, ClubName, Description, AdvisorID FK → FACULTY
MEMBERSHIP	StudentID PK/FK, ClubID PK/FK, JoinDate
CLUB_OFFICER	ClubID PK/FK, StudentID PK/FK, Position, StartDate, EndDate
EVENT	EventID PK, ClubID FK, EventName, EventDate, Description
ATTENDANCE	StudentID PK/FK, EventID PK/FK, AttendanceStatus
ROOM	RoomID PK, Building, RoomNumber, Capacity
RESERVATION	ReservationID PK, EventID FK, RoomID FK, StartTime, EndTime
CLUB_BUDGET	BudgetID PK, ClubID FK, BudgetYear, Amount
EXPENSE	ExpenseID PK, ClubID FK, ExpenseDate, Amount, Description

Relationships

- Faculty 1:N Club
- Student M:N Club through Membership
- Club 1:N Event
- Student M:N Club through ClubOfficer; Position and dates are attributes.
- Student M:N Event through Attendance
- Event 1:N Reservation; Room 1:N Reservation
- Club 1:N ClubBudget
- Club 1:N Expense



Design decision

Budget is stored in CLUB_BUDGET with BudgetYear instead of one Budget field in CLUB. This preserves historical and year-specific budgets.

Example queries in English (not SQL)

- Find all students who are officers in the Computer Science Club.
- List all events scheduled for next week with their room reservations.
- Find all expenses made by each club during the current academic year.